AKTU

B.Tech 6th Sem

CS IT & CS Allied

Software Engineering

Unit-1: One Shot

Gateway Classes

Syllabus: Unit - I

Introduction: Introduction to Software Engineering, Software Components, Software Characteristics, Software Crisis, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes, Software Quality Attributes. Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Models

Gateway Classes

Software + Engineering

Software Engineering is an engineering branch related to the evolution of software product using well-defined scientific principles, techniques, and procedures. The result of software engineering is an effective and reliable software product.

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Software Components

Software components are the building blocks of a software system. They interact to perform specific tasks. The primary components are:

1. Programs: Code is written to perform specific functions.
2. Data: Information used or processed by the program.
3. Documentation: Manuals, help files, and guides to assist users and developers.
4. Interfaces: Mechanisms through which components communicate.

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Software Characteristics

AKTU 2023-24, 2021-22

1. Intangibility: Software cannot be physically touched or seen.
2. Flexibility: It is easily modifiable compared to physical products.
3. Reliability: Software should function correctly under specified conditions.
4. Maintainability: It must be easy to modify, debug, and upgrade.
5. Scalability: Software can handle increased loads or requirements.
6. Portability: It can run across different hardware or operating systems with minimal changes.

2+3

5+9=?

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Software Engineering required....

- To manage Large software
- For more Scalability
- Cost Management
- To manage the dynamic nature of software
- For better quality Management

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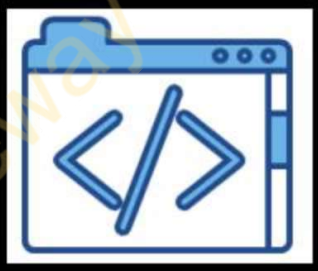
Importance of Software Engineering

- Reduces Complexity
- Minimize Software cost
- Decrease time Handling
- Big Project
- Effectiveness
- Reliable software



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Program vs Software

Program:

- Program is a combination of source code & object code.

Software:

- Software is a set of instructions, data or programs used to operate computers and execute specific tasks.

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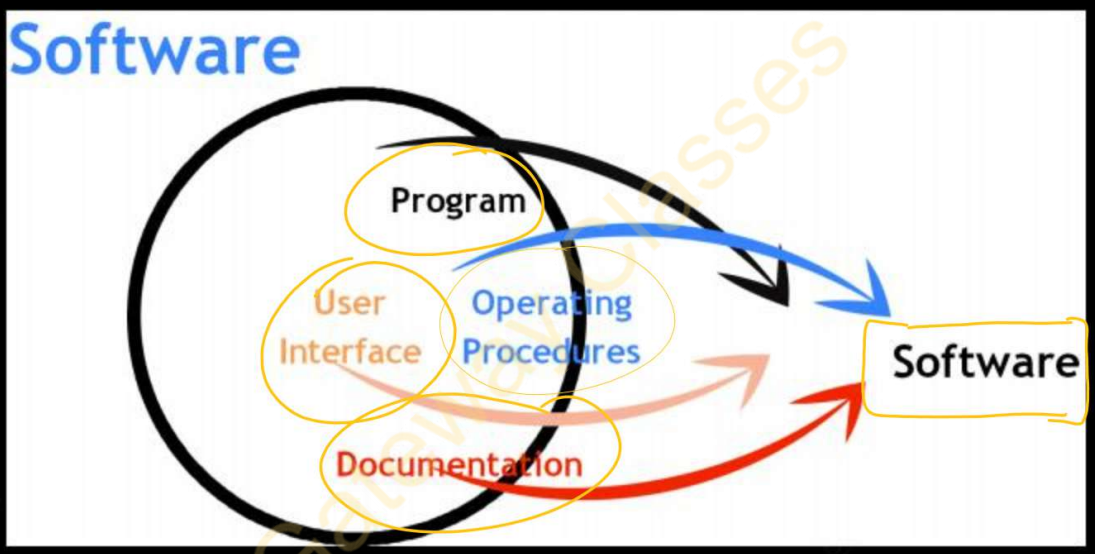
Software


Software
Program
User Interface
Operating Procedures
Documentation
Software

Software Crisis

Software crisis is a set of difficulties or problems encountered while developing software.

1. Size: Software is becoming more expensive and more complex with the growing complexity and expectation out of software. For example, the code in the consumer product is doubling every couple of years.

2. Quality: Many software products have poor quality, i.e., the software products defects after putting into use due to ineffective testing technique. For example, Software testing typically finds 25 errors per 1000 lines of code.

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Software Crisis

3. Cost: Software development is costly i.e. in terms of time taken to develop and the money involved. For example, Development of the FAA's Advanced Automation System cost over $700 per lines of code.

4. Delayed Delivery: Serious schedule overruns are common. Very often the software takes longer than the estimated time to develop, which in turn leads to cost shooting up. For example, one in four large-scale development projects is never completed.

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Software Engineering Processes

AKTU 2022-23

- A software engineering process is a set of activities performed to design, develop, test, and maintain software. These processes typically include:

Requirement Analysis: Understanding user needs.

System Design: Creating architectural and detailed designs.

Implementation: Writing code.

Testing: Ensuring the software meets requirements.

Deployment: Releasing the software for use.

- Maintenance: Updating and fixing software post-deployment.

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Similarity and Differences from Conventional Engineering Processes

Similarities:

- Both follow systematic approaches to problem-solving.
- Require thorough planning and design phases.
- Emphasize quality assurance and testing.

Differences:

Aspect	Software Engineering	Conventional Engineering
Nature of Product	Intangible, logical design	Tangible, physical structures
Changeability	Easily modifiable	Expensive and difficult to modify
Tools	Programming languages, IDEs	CAD tools, physical materials
Failures	Bugs, logical errors	Physical failures, wear and tear

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Software Quality Attributes (AKTU-2023-24)

These are the characteristics that define the quality of a software system:

- 1. Functionality: Performs required tasks correctly.
- 2. Reliability: Operates consistently without failure.
- 3. Usability: Easy to use and understand.
- 4. Efficiency: Optimally uses system resources.
- 5. Maintainability: Easy to update and debug.
- 6. Portability: This can be used across various platforms.

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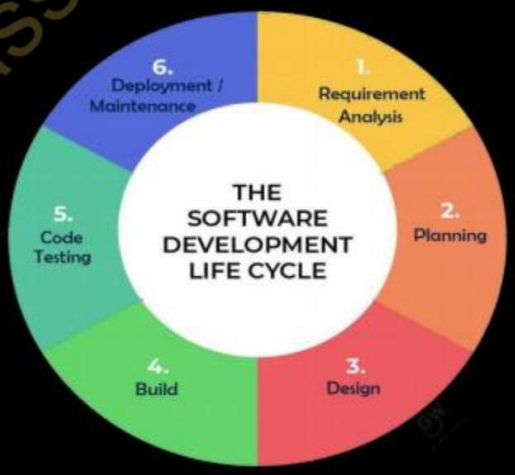
SDLC Models

- A software Development life cycle model is a pictorial and diagrammatic representation of the software life cycle.
- SDLC is used by the software industry to design, develop and test high quality softwares.



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SDLC Models
(Software Development Life Cycle)

SDLC Models

Phase 1: Requirement Analysis:

- Gathering business requirement Creating process diagrams Performing a detailed analysis

Phase 2: Planning

- Identification of the system for development Feasibility assessment
- Creation of project plan

Phase 3: Design

- This phase includes business rules, pseudo-code, screen layouts, and other necessary documentation.
- Designing of software model.

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Phase 4: Build/Development:

- The implementation of design begins concerning writing code. Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

Phase 5: Code Testing Writing test cases Execution of test cases

- Tester follows the Software Testing Life Cycle activities to check the software for errors, bugs, and defects.

Phase 6: Deployment/Maintenance

- In Deployment software is deployed to a real-life environment where the actual user begins to operate the software.
- Support the software users. Software maintenance.
- Software changes and adjustment

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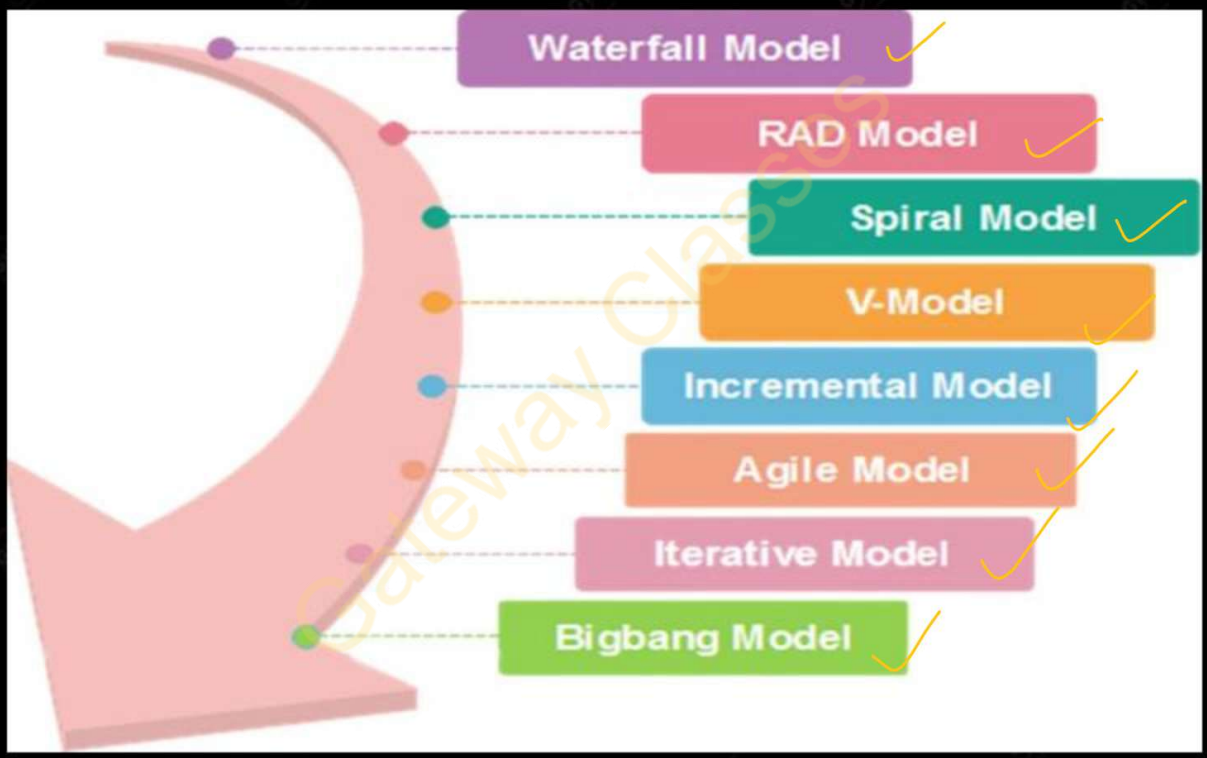
SDLC Models



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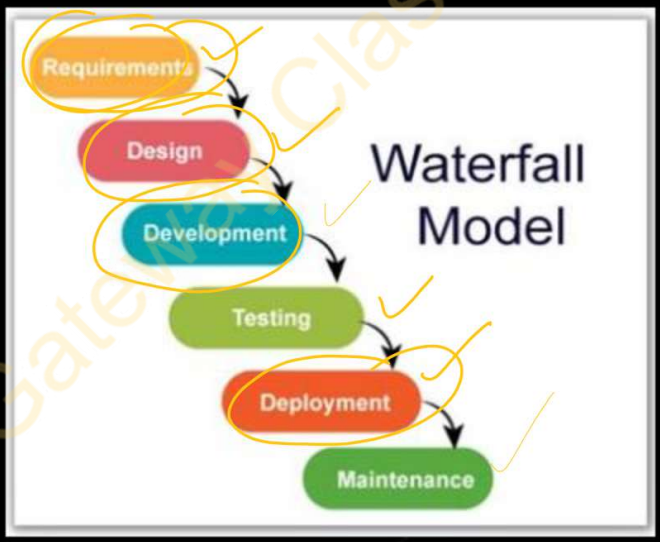
Waterfall model (AKTU-2022-23)

- In Waterfall Model progress is seen as flowing steadily downwards (like a waterfall).



SDLC

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Waterfall model advantages & Disadvantages

Advantages:

- it is easy to use and understand.
- No overlapping between the phases. This model is suitable for small projects.
- Provides better documentation for the employees. Low Cost
- It is easy to maintain.

Disadvantages:

- This model does not work for large projects.
- We cant not go back at previous phase to change anything or any requirement.
- If customer is not satisfied with the project then it is hard to change the requirements. High risk to make a project.

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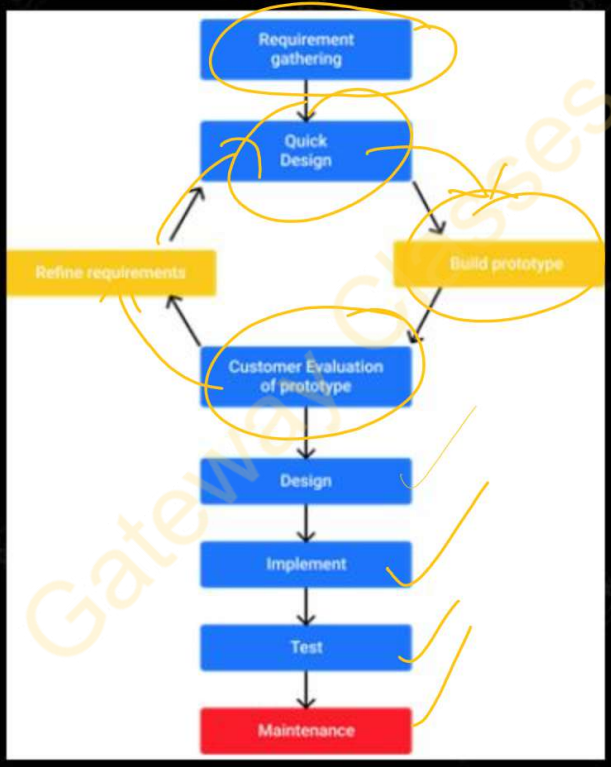
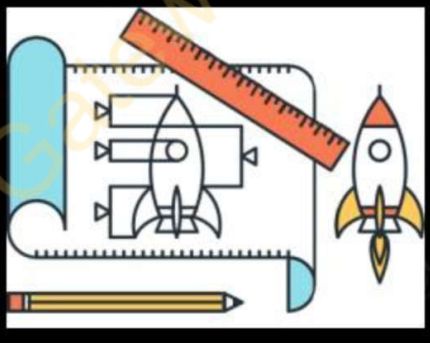
Prototype model (AKTU-2021-22)



Prototype model

- The prototype model requires that before carrying out the development of actual software, a working prototype of the system should be built. A prototype is a toy implementation of the system. A prototype usually turns out to be a very crude version of the actual system, possible exhibiting limited functional capabilities, low reliability, and inefficient performance as compared to actual software. In many instances, the client only has a general view of what is expected from the software product.



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Prototype model advantages & Disadvantages

Advantages:

- It provides higher customer satisfaction and get reviews from the customers. Requirement can change easily according to the customer. It means this model has high adaptability.
- It gives high flexibility.
- Prototype model increase the involvement of customer. Reduce Maintenance cost.

Disadvantages:

- Difficult to know how long the project will last. Prototyping tools are expensive.
- If a customer refuses to accept the prototype, then there may be wastage of a lot of cost and developer's efforts.
- It is a time-consuming process.

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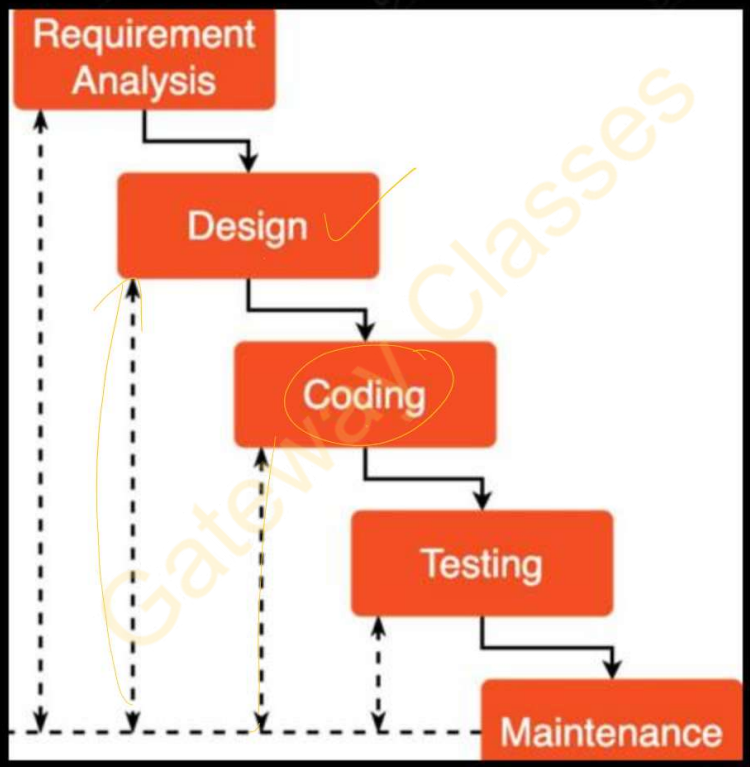
Iterative Waterfall model



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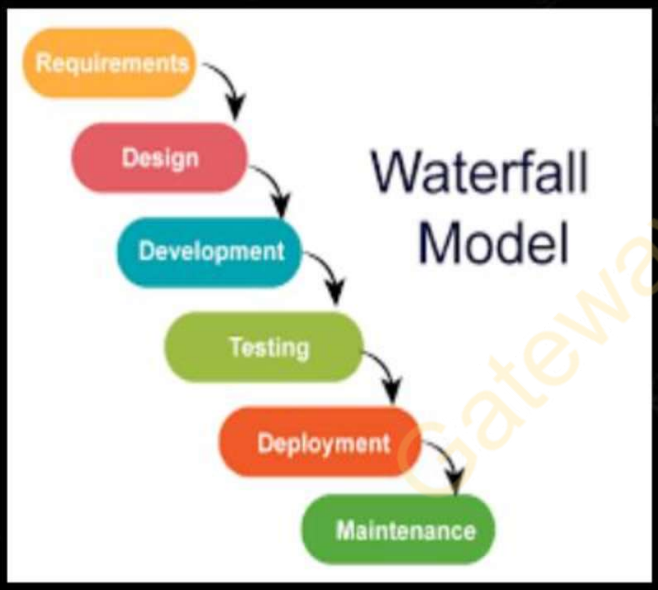
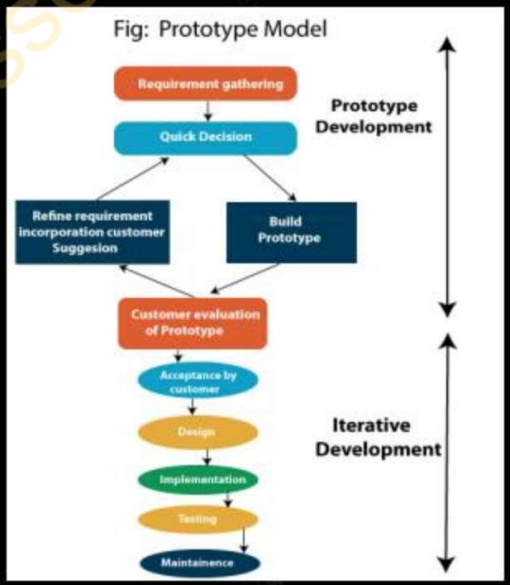
Spiral Model

Spiral model is the combination of Waterfall Model and Prototype Model





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Spiral model

- "Spiral model is the Risk Driven model. This model is best used for large projects which involves continues enhancements."

- The spiral model was first mentioned by Barry Boehm in his 1986 paper.

- Each phase in spiral model begins with a design goal and ends with the client reviewing the progress.

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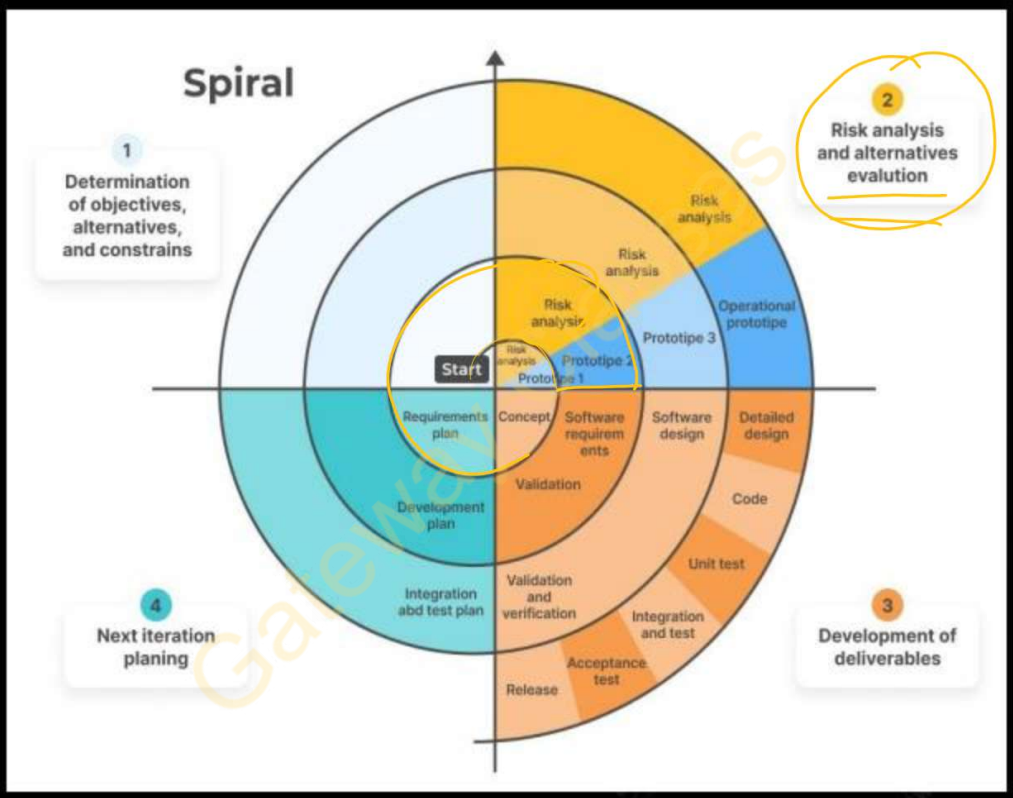
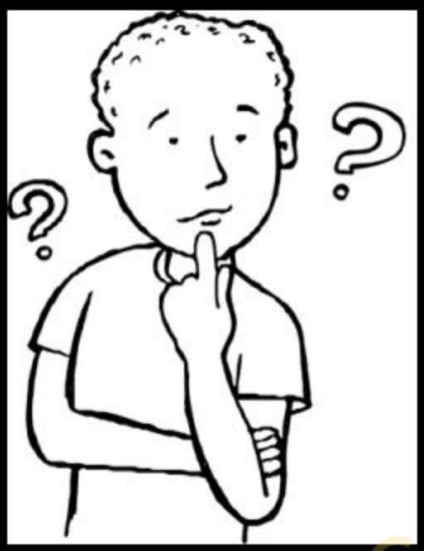

Spiral
1
Determination of objectives, alternatives, and constrains
2
Risk analysis and alternatives evalution
3
Development of deliverables
4
Next iteration planing
Start
Risk analysis
Prototipe 1
Risk analysis
Risk analysis
Prototipe 3
Operational prototipe
Requirements plan
Concept
Software requirem ents
Software design
Detailed design
Validation
Development plan
Code
Unit test
Integration abd test plan
Validation and verification
Integration and test
Acceptance test
Release

When to use Spiral Model?



- When deliverance is required to be frequent.
- When the project is large
- When requirements are unclear and complex
- When changes may require at any time Large and high budget projects

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Spiral model advantages & Disadvantages

Advantages:

- High amount of risk analysis.
- Useful for large and mission-critical projects.

Disadvantages:

- Can be a costly model to use.
- Risk analysis needed highly particular expertise. Doesn't work well for smaller projects.

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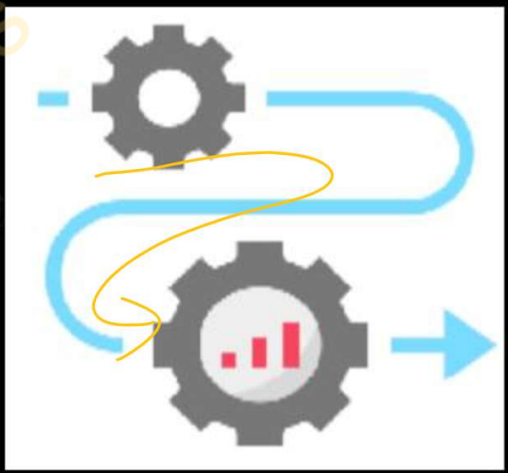
Evolutionary Development Model

- Evolutionary models are iterative type models which allow to develop more complete version of the software.
- In this model, the system is first broken down into the several modules or functional units that can be incrementally implemented and delivered.
- The developer first develop the core modules of the system.
- This model is also known as the successive versions model.



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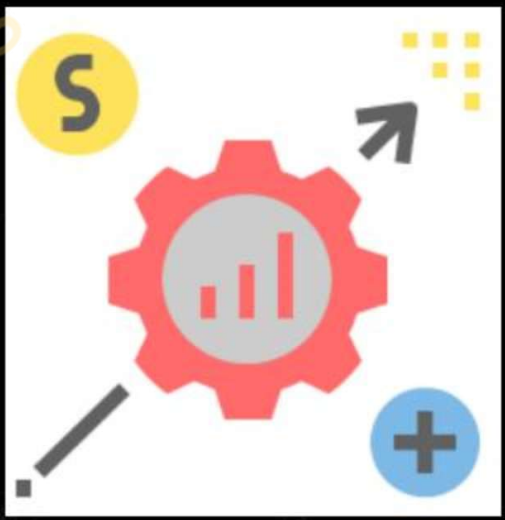
Evolutionary Development Model Advantages & Disadvantages

Advantages:

- In evolutionary model, a user gets a chance to experiment partially developed system.
- It reduces the error because the core modules get tested thoroughly.

Disadvantages:

- Sometimes it is hard to divide the problem into several versions that would be acceptable to the customer which can be incrementally implemented and delivered.



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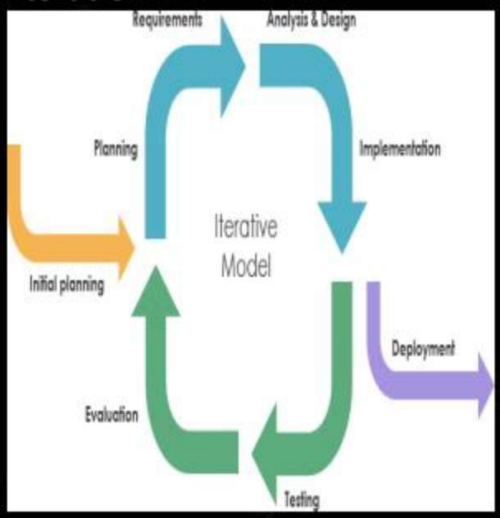
Iterative Enhancement Model

- In this Model, you can start with some of the software specifications and develop the first version of the software. After the first version if there is a need to change the software, then a new version of the software is created with a new iteration. Every release of the Iterative Model finishes in an exact and fixed period that is called iteration.



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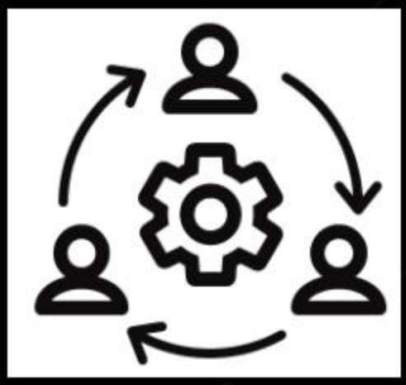
Iterative Enhancement Model Advantages & Disadvantages

Advantages:

- Testing and debugging during smaller iteration is easy. A Parallel development can plan.
- Risks are identified and resolved during iteration. Limited time spent on documentation and extra time on designing.

Disadvantages:

- It is not suitable for smaller projects. More Resources may be required.
- Requirement changes can cause over budget.
- Project completion date not confirmed because of changing requirements.



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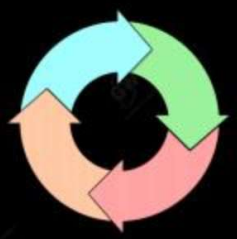
Product vs Process

Product:

- A product may be defined as an output. A product can produce the good services.

Process:

- A process may be defined as an investment of time and efforts for getting a good product.



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Thank You